

DIEGO MAGANA

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Principal Engineer building production ML and data systems since 2017: forecasting, pricing, and classification models, Azure data platforms, and LLM evaluation pipelines. Owns ML work end to end, from data and modeling through deployment and evaluation.

EDUCATION

Georgia Institute of Technology , M.S. Computer Science (Artificial Intelligence)	<i>Expected May 2029</i>
Amherst College , B.A. · NCAA D-III Basketball, NESCAC Champion, NCAA Semifinalist	<i>2017</i>

EXPERIENCE

Bravon Industries , Irvine, CA	Jan '23 – Present
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<i>Principal Engineer (Contract)</i>	<i>Oct '25 – Present</i>
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- Built a RAG evaluation harness benchmarking open-weights LLMs (Llama 3.1, Mistral) against managed copilots, scoring retrieval, faithfulness, and numerical accuracy as separate gates to inform build vs. buy.
- Added a CI regression gate and per-component tracing that localizes failures to retrieval or generation.

<i>Chief Information Officer</i>	<i>Mar '24 – Oct '25</i>
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- Led Azure cloud and data architecture as the sole in-house technical lead, owning the production AL codebase and directing contractors and an outsourced engineering team.
- Integrated Business Central (ERP) with trading-partner and operational systems across finance, supply chain, and EDI.

<i>Principal Engineer (Contract)</i>	<i>Jan '23 – Mar '24</i>
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- Architected the company's migration from Odoo to the Microsoft stack on Azure and built the cloud data platform and ML/analytics infrastructure still in production.
- Built ETL pipelines and REST API integrations feeding the production data platform.

Mavco LLC , Scottsdale, AZ	Aug '17 – Jan '23
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<i>Senior Data Scientist</i>	<i>Aug '20 – Jan '23</i>
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- Built the company's first ML function and owned the full lifecycle of its forecasting, pricing, and classification models, from data engineering and modeling through production deployment.
- Drove a ~10-point lift in adjusted gross margin through these models; outputs informed seven-figure capital allocation.
- Scaled the modeled catalog from hundreds of SKUs to tens of thousands with automated pipelines that assessed hundreds of thousands of candidate products.

<i>Data Scientist</i>	<i>Aug '17 – Aug '20</i>
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- Built ML pipelines for sales and pricing prediction, plus financial models used in executive planning.

PROJECTS

Transformer Interpretability github.com/diego-magana/gpt	<i>Python, PyTorch</i>
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- Trained a character-level transformer and built caching, ablation, and activation-patching tooling from scratch, tracing its next-token prediction to one attention head (ablating it raised loss 7× more than any other head).
- Confirmed the mechanism with a redundancy test and independently in weight space (QK circuit analysis), then replicated it across five training seeds: the mechanism is robust, the head's position an initialization artifact.

Neural Networks from First Principles github.com/diego-magana	<i>Python, PyTorch</i>
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- Built a reverse-mode autodiff engine and a progression of character-level language models from bigram to WaveNet (2.01 validation NLL), verifying hand-written backprop against PyTorch autograd.
- Packaged the autodiff engine as a tested library with CI and extended it beyond the reference implementation (additional activations, multi-class classification).

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++

Machine Learning: PyTorch, scikit-learn, transformers, forecasting, pricing and classification models, model evaluation

LLMs: RAG, LLM evaluation (retrieval, faithfulness, accuracy), open-weights models (Llama, Mistral), interpretability

MLOps & Data: Docker, GitHub Actions (CI/CD), Modal, W&B, Azure (AI Foundry, data platforms), ETL, REST APIs